A Cross-Domain Stability Metric for Cardiac Arrhythmia Detection and Physiological Instability Monitoring

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Abstract

Physiological monitoring methods for detecting rhythm instability typically rely on signal-specific feature pipelines, supervised classifiers trained on labeled examples, or modality-specific scoring rules that do not transfer across measurement domains. This paper evaluates a stability metric requiring no supervised model training, $\Phi = I \times \rho - \alpha \times S$, computed from physiological time-series data using domain-appropriate observable mappings. The same three-term structure has been previously evaluated in the author's portfolio on mechanical, electrical, aerospace, computational, geophysical, and quantum datasets, most commonly with $\alpha = 0.1$. The present paper reports its application to cardiac and neural physiological signals across eight validation tests on publicly available PhysioNet datasets. On the MIT-BIH Arrhythmia Database, under per-record threshold calibration, Φ discriminated arrhythmia from normal sinus rhythm at AUC 0.9148 across 1,022 evaluable 60-second windows from 34 records, with a shuffle audit AUC of 0.5003 confirming the signal was not an artifact of label leakage. On the MIT-BIH Atrial Fibrillation Database, Φ produced AUC 0.5556 across 11,988 windows, a result interpreted as a scope boundary supporting the structural reading that Φ tracks transitions rather than classification between stable rhythm modes. Resolution scaling on the same cardiac dataset produced AUC 0.8775 at 30-second windows and AUC 0.6376 at 5-second windows, indicating graceful degradation rather than abrupt collapse. A head-to-head comparison against 16 standard heart-rate-variability metrics on 811 matched windows placed Φ at AUC 0.9015, sixth among 16 metrics, behind cardiac-specific RMSSD (0.9706) by 0.0691 AUC. Exploratory neural validation on the CHB-MIT Scalp EEG Database produced weak and direction-dependent window-level results across four tested patients, with tested configurations generally near chance to modestly above chance and requiring patient-specific interpretation. A 3-consecutive-hit audit on chb01 falsified an earlier reported 71% event-level seizure detection result by reducing detection to 0 of 7 while leaving window-level AUC unchanged at 0.5654, consistent with a chance-rate calculation that gave approximately 78% probability of at least one chance hit per seizure under the original single-hit criterion. A K-of-N event-detection rule (3 hits in any 10 windows) increased single-patient event sensitivity to 67% (4 of 6 seizures with sufficient coverage) at 0.74 false alarms per hour. Results are bounded to the tested datasets, the tested observable mappings, and the tested configurations. Generalization beyond these specific choices, including to other physiological modalities, to multi-patient validated seizure prediction, or to actual wearable hardware deployment, is not established by the evidence reported here.

1 Introduction

Physiological monitoring systems generate continuous time-series data from sensors measuring cardiac activity, brain electrical activity, blood oxygenation, respiration, and other physiological variables. The detection of instability or transition toward failure in these systems is operationally important across clinical bedside monitoring, ambulatory monitoring, and wearable consumer devices. The methods commonly used to perform this detection are typically signal-specific. Cardiac monitoring uses heart-rate-variability metrics computed from inter-beat intervals (Shaffer and Ginsberg, 2017). Electroencephalography monitoring uses spectral analysis, channel-specific morphology features, or supervised classifiers trained on labeled seizure data (Rasheed et al., 2020). Each of these methods is tuned to the specific physiological modality and does not transfer to other modalities without redesign of the feature pipeline.

This paper evaluates whether a single stability metric requiring no supervised model training, computed from physiological time-series data using domain-appropriate observable mappings, detects instability across cardiac and neural signals under one structural formulation. The metric is defined as

$$\Phi = I \times \rho - \alpha \times S, \tag{1}$$

where I is an identity component computed from a baseline-referenced quality measure, ρ is a temporal coherence component computed from autocorrelation of the signal, S is a disorder component computed from Shannon entropy of a derived distribution, and α is a coupling constant set to 0.1 for cardiac tests, with exploratory EEG analyses testing higher entropy weighting at $\alpha = 0.2$ and $\alpha = 0.55$. The same three-term structure has been previously evaluated in the author's portfolio on mechanical bearing degradation, turbofan engine degradation, electrical power-grid frequency excursions, retrospective seismic strain windows, neural-network training trajectories, and quantum hardware qubit calibration data, most commonly with $\alpha = 0.1$, with documented exceptions including $\alpha = 0.034$ for battery data and a normalized entropy formulation for class-count-agnostic neural-network operation. The present paper extends the evaluation to physiological time-series data with the same three-term structure and reports both the strong cardiac result and the weaker exploratory neural result.

The paper addresses four questions. First, does Φ discriminate arrhythmia from normal sinus rhythm on a publicly available cardiac arrhythmia dataset under per-record threshold calibration, and does the result survive a shuffle-label audit? Second, does Φ classify atrial fibrillation against normal sinus rhythm on a publicly available atrial fibrillation dataset, and what does the result say about the scope of what Φ measures? Third, does Φ on cardiac data degrade gracefully across a range of window lengths from longer clinical-style windows to shorter low-latency windows, and how does Φ compare against standard heart-rate-variability metrics on the same task? Fourth, does the same formula structure transfer to neural electroencephalography signals for seizure prediction, and what are the boundary conditions on that extension?

The contributions are as follows. The paper reports a primary cardiac validation layer covering arrhythmia detection, an atrial-fibrillation negative-control boundary result, resolution scaling across four window lengths, and a head-to-head comparison against 16 standard heart-rate-variability metrics. The paper also reports an exploratory neural validation layer with explicit patient-specific direction caveats and explicit single-patient bounds on the K-of-N event-detection result. The paper additionally reports a methodological self-falsification result showing that a permissive single-hit EEG event-detection rule produced an apparent seizure-detection result consistent with chance accumulation across many independent windows, and that a 3-consecutive-hit audit reduced the event-level result to 0 of 7 while leaving window-level AUC unchanged; this audit motivated the K-of-N event-detection rule used in the corrected neural analysis. All evidence is from publicly available PhysioNet datasets with published annotations, including cardiologist beat annotations for MIT-BIH. No synthetic data is used.

The methodological self-falsification is reported as follows. An earlier reported event-level seizure detection result of 71% on chb01 was reduced to 0 of 7 under a 3-consecutive-hit audit, with window-level AUC unchanged at 0.5654. A chance-rate calculation, $1 - 0.99^{150} \approx 78\%$, accounts for the original number under the assumption of random chance accumulation across the approximately 150 preictal windows per seizure. The K-of-N event-detection rule (3 hits in any 10 windows) was introduced to increase event sensitivity for scattered threshold-hit patterns without the false-positive accumulation problem of single-hit criteria. The K-of-N result is reported as single-patient exploratory event-level evidence, not as a validated cross-patient seizure prediction claim.

Results are bounded to the specific datasets, observable mappings, configurations, and audit protocols used. Generalization beyond these specific choices, including to other cardiac patient populations, to other physiological modalities such as photoplethysmography or respiration, to multi-patient validated seizure prediction, or to deployment on actual wearable hardware, is not established by the evidence reported here.

2 Related Work

Cardiac arrhythmia detection has a long history grounded in heart-rate variability analysis. Standard time-domain metrics include the standard deviation of normal-to-normal intervals (SDNN), the root mean square of successive differences (RMSSD), and the percentage of successive intervals differing by more than 50 milliseconds (pNN50). Standard frequency-domain metrics include low-frequency and high-frequency spectral power and their ratio. Standard nonlinear metrics include sample entropy and Poincaré plot descriptors such as SD1 and SD2 (Shaffer and Ginsberg, 2017; Malik et al., 1996). These metrics are tuned to cardiac inter-beat structure and generally require reinterpretation or replacement when applied outside cardiac interval analysis.

Atrial fibrillation detection methods commonly operate as classifiers between AFib and normal sinus rhythm using rhythm-pattern matching, RR-interval irregularity statistics, beat morphology features, or supervised machine learning trained on labeled rhythm segments (Petrenas et al., 2015; Faust et al., 2018). These methods are commonly framed as rhythm-state classification methods in which AFib is treated as a rhythm pattern distinguishable from normal sinus rhythm under sustained observation. The present paper reports a negative result on AFib classification using a stability metric designed for transition detection, which is consistent with the structural difference between mode classification and transition tracking.

Seizure prediction from scalp electroencephalography is an active research area covering both feature-based methods and deep-learning methods. Feature-based methods commonly use spectral band power, spike-rate analysis, phase synchrony between channels, or graph-theoretic measures computed across multi-channel recordings (Rasheed et al., 2020; Kuhlmann et al., 2018). Deep-learning methods commonly use convolutional, recurrent, attention-based, or hybrid architectures trained on labeled preictal segments. Many seizure-prediction systems require patient-specific calibration because preictal physiology can vary by seizure focus, age, and underlying pathology. The present paper reports exploratory EEG results that are consistent with this patient-specific requirement.

A related line of work in the author's prior portfolio studies cross-domain stability monitoring under a single equation structure. Prior technical reports by the present author have evaluated a three-term stability metric on mechanical bearing degradation, turbofan engine degradation, electrical power-grid frequency excursions, retrospective seismic strain windows from the USGS Donna Lea strainmeter, neural-network training trajectories, and quantum hardware qubit calibration data on IBM Quantum backends (Barnicle, 2026a,b,c). Most of those evaluations used a coupling constant of 0.1, with documented exceptions including a coupling constant of 0.034 for battery degradation data and a normalized entropy formulation for class-count-agnostic operation on neural-network architectures spanning 2, 10, and 100 output classes. The present paper extends that line of work to physiological time-series data using the same three-term structure with cardiac-specific and neural-specific observable mappings.

The methodological audit reported in the present paper, in which an earlier event-level seizure detection result was reduced from 71% to 0 of 7 under a 3-consecutive-hit requirement with window-level AUC unchanged, is consistent with the general statistical problem of chance accumulation under repeated testing or repeated window evaluation (Benjamini and Hochberg, 1995). The K-of-N event-detection rule used in the corrected analysis is a multi-window confirmation approach with a coverage requirement and a refractory period.

3 Methods

3.1 The Stability Metric

The stability metric Φ is a scalar computed from three components:

$$\Phi = I \times \rho - \alpha \times S, \tag{2}$$

where I is a baseline-referenced quality component, ρ is a temporal coherence component, S is a disorder component, and α is a coupling constant. The components are computed from a windowed segment of physiological signal data using domain-appropriate observable mappings. The same formula structure is used for cardiac and neural physiological signals; the observable mappings differ. Each test fixes the formula structure, the observable mapping, and the coupling constant before execution for that test.

3.2 Cardiac Observable Mappings

For cardiac signals, the input to Φ is a sequence of inter-beat (RR) intervals derived from electrocardiogram beat annotations. Within a window of duration W seconds containing at least *min_rr* RR intervals, the cardiac mappings are as follows.

The identity component is defined as the ratio of the baseline RR-interval standard deviation to the current window's RR-interval standard deviation, clamped to $[0, 1]$:

$$I = \mathrm{clip}_{[0,1]}\left(\frac{\sigma_{\mathrm{RR}}^{\mathrm{baseline}}}{\max(\epsilon, \sigma_{\mathrm{RR}}^{\mathrm{current}})}\right), \tag{3}$$

where $\sigma_{\mathrm{RR}}^{\mathrm{baseline}}$ is the median per-window standard deviation across windows labeled as NORMAL-compatible for the same record, $\sigma_{\mathrm{RR}}^{\mathrm{current}}$ is the standard deviation of the RR intervals within the current window, and ϵ is a small positive constant used to avoid division by zero. The clamping bounds I to $[0, 1]$ and prevents low-variability windows from producing values above one.

The temporal coherence component is defined as the lag-1 Pearson autocorrelation of the RR-interval sequence within the window, clamped to $[0, 1]$:

$$\rho = \mathrm{clip}_{[0,1]}\big(r_1(\mathrm{RR})\big), \tag{4}$$

where r_1 denotes the Pearson correlation computed over the lag-1 paired observations $(\mathrm{RR}[i], \mathrm{RR}[i+1])$ for $i = 0, \ldots, n-2$.

The disorder component is defined as the Shannon entropy in bits of a histogram of the RR-interval distribution within the window:

$$S = -\sum_k p_k \log_2 p_k, \tag{5}$$

where p_k are the bin probabilities of the histogram, computed using Freedman-Diaconis bin selection. The Freedman-Diaconis rule sets the bin width as $2 \times \mathrm{IQR} \times n^{-1/3}$, where IQR is the interquartile range of the RR sequence and n is the number of intervals in the window. When the IQR is zero or the resulting bin count is degenerate, the implementation falls back to $\lceil \sqrt{n} \rceil$ bins.

For cardiac tests, the coupling constant is fixed at $\alpha = 0.1$.

3.3 Neural Observable Mappings

For neural signals, the input to Φ is a multi-channel scalp EEG recording sampled at 256 Hz. Within a window of duration W seconds, the neural mappings are as follows.

The identity component is defined as the ratio of the baseline RMS amplitude to the current window's RMS amplitude, clamped to $[0, 1]$:

$$I = \mathrm{clip}_{[0,1]}\left(\frac{A_{\mathrm{RMS}}^{\mathrm{baseline}}}{\max(\epsilon, A_{\mathrm{RMS}}^{\mathrm{current}})}\right), \tag{6}$$

where $A_{\mathrm{RMS}}^{\mathrm{baseline}}$ is computed from clean non-seizure files for the same patient after per-channel demeaning, per-channel z-score normalization, and samplewise averaging across channels, and $A_{\mathrm{RMS}}^{\mathrm{current}}$ is the RMS amplitude of the resulting channel-aggregated signal within the current window.

The temporal coherence component is defined as the lag-ℓ Pearson autocorrelation of the channel-aggregated EEG signal, with ℓ corresponding to a 1.0-second lag at the 256 Hz sampling rate. ρ is clamped to $[-1, 1]$ for the EEG mapping; unlike the cardiac mapping, the EEG implementation preserves the sign of the autocorrelation rather than zeroing negative values.

Channel aggregation is performed as follows. Each channel is demeaned and z-score normalized over the loaded record. The channel-aggregated signal is the mean across normalized channels at each time sample. RMS amplitude in the identity component above and autocorrelation in the coherence component above are computed on this channel-aggregated signal. This aggregation order, per-channel normalization followed by averaging across channels and then per-window RMS, is used to reduce the influence of channel-scale differences on the per-window quantities.

The disorder component for neural signals is the absolute deviation of the current window's spectral entropy from a per-patient baseline spectral entropy:

$$S_{\text{dev}} = |S_{\text{current}} - S_{\text{baseline}}|, \tag{7}$$

where S_{current} is the Shannon entropy in bits of the normalized power spectrum of the channel-aggregated signal in the current window, computed from the squared magnitude of the discrete Fourier transform with the DC bin removed and the remaining bins normalized to a probability distribution. S_{baseline} is the median value of S_{current} across clean interictal windows for the same patient. Spectral entropy is used for the exploratory EEG mapping because this implementation evaluates disorder through the signal's frequency-domain power distribution rather than through the amplitude-value distribution used for cardiac RR intervals. The deviation form is used because raw EEG spectral entropy is patient-specific, and the scientific question for the neural tests is whether spectral disorder moves away from each patient's own baseline during preictal windows.

For neural tests, the coupling constant is set to $\alpha = 0.2$ for the K-of-N event-detection test on chb01 and to $\alpha = 0.55$ for the multi-patient optimization across chb01, chb02, chb03, and chb04. This EEG observable mapping is intentionally simple and is treated as exploratory; it does not use spectral band power, channel-localized onset features, or phase-synchrony features.

3.4 Baseline Threshold Calibration

In the cardiac tests, the threshold Φ_c used for window-level instability classification is calibrated per record. In the neural tests, Φ_c is calibrated per patient from clean interictal baseline windows. For each cardiac record (or each EEG patient), all baseline-compatible windows produce a distribution of Φ values. The threshold Φ_c is set to the quantile of that distribution corresponding to a target false-positive rate of 0.01 for cardiac tests and for the original chb01 EEG window-level audit, and 0.05 for the K-of-N neural event-detection test.

For cardiac tests where lower Φ indicates instability, and for the primary chb01 and chb04 lower-direction EEG analyses, a window is classified as instability-positive if $\Phi < \Phi_c$. For EEG direction analysis, higher-direction and absolute-deviation variants are also evaluated diagnostically to assess patient-specific preictal directionality.

Per-record (cardiac) and per-patient (neural) calibration is used in this paper because the entropy component S is computed in bits rather than normalized to $[0, 1]$, and because patient-to-patient variability in baseline Φ values is substantial. Cardiac records with no usable NORMAL-compatible windows are excluded from per-record calibration. For the primary cardiac arrhythmia detection test, this exclusion criterion yields 34 of 48 records and 1,022 evaluable 60-second windows. For the HRV comparison test, a stricter inclusion rule requiring at least five NORMAL-compatible windows per record is used so that all 16 compared HRV metrics and Φ are evaluated on the same matched window set with sufficient per-record baseline support; this stricter rule yields 811 matched evaluable windows. For neural event-level evaluation, the K-of-N rule is evaluated only for seizures with at least N valid preictal windows under the coverage guard described below.

The patent specification corresponding to this work codifies both fixed-threshold and calibrated-threshold approaches as configurable embodiments; the present paper uses calibrated thresholds throughout the physiological tests.

3.5 Window-Level and Event-Level Detection

Window-level detection produces a binary instability flag per window based on whether $\Phi < \Phi_c$ for that window in the lower-direction tasks. Window-level performance is reported using the area under the receiver operating characteristic curve (AUC), precision, recall, specificity, F1 score, and Matthews correlation coefficient (MCC). The AUC is computed from a continuous Φ-derived score with the sign adjusted according to the tested direction of instability, rather than from the binary threshold flags. Thus, AUC is independent of the particular Φ_c selected and reflects the underlying separability of the two classes. For neural tests where direction varies by patient, both lower-direction and higher-direction diagnostics are reported.

Event-level detection aggregates window-level flags into event predictions for downstream evaluation. Two event-detection rules are evaluated in this paper. The 3-consecutive-hit rule produces an event prediction when at least three consecutive windows fall below Φ_c. The K-of-N rule (with $K = 3$, $N = 10$) produces an event prediction when at least K windows within any sliding block of N consecutive windows fall below Φ_c, with the block sliding forward by one window per step. The K-of-N rule is paired with a refractory period of 1,800 seconds (thirty minutes) during which subsequent threshold crossings do not produce additional event predictions, and a coverage guard requiring at least N valid windows within the relevant analysis period before the rule is evaluated.

Event-level performance is reported as seizure sensitivity (the fraction of annotated seizures with at least one event prediction in the preictal window) and false alarm rate (event predictions per hour during clean interictal periods).

3.6 Heart-Rate-Variability Comparison

For the cardiac HRV comparison, Φ was evaluated against 16 standard heart-rate-variability metrics computed on the same MIT-BIH arrhythmia windows. The comparison included time-domain metrics (SDNN, RMSSD, pNN50, SDSD, mean heart rate), frequency-domain metrics (low-frequency power, high-frequency power, LF/HF ratio), and nonlinear metrics (sample entropy, DFA α_1, SD1, SD2, SD1/SD2). The HRV comparison used the stricter inclusion rule of at least five NORMAL-compatible windows per record described in the previous subsection, yielding 811 matched evaluable windows. All metrics were evaluated on the same labels and under matched per-record calibration conditions. The comparison is interpreted as a matched benchmark of a cross-domain stability metric against cardiac-specific HRV features, not as a claim that Φ is the best cardiac-specific detector.

3.7 Datasets

MIT-BIH Arrhythmia Database. The cardiac arrhythmia tests use the MIT-BIH Arrhythmia Database (Moody and Mark, 2001; Goldberger et al., 2000), available from PhysioNet. The database contains 48 ambulatory ECG recordings of approximately 30 minutes each, sampled at 360 Hz, with cardiologist beat annotations covering approximately 110,000 beats. Beat symbols are grouped into NORMAL-compatible ({L, N, R, e, j}) and ARRHYTHMIA ({A, E, F, J, Q, S, V, a}) classes according to the implementation's symbol mapping. RR intervals are derived from the beat annotation timestamps. The records used in the primary test are those with at least one window of NORMAL-compatible beats available for per-record threshold calibration, which yields 34 of 48 records and 1,022 evaluable 60-second windows.

MIT-BIH Atrial Fibrillation Database. The atrial fibrillation tests use the MIT-BIH Atrial Fibrillation Database (Moody and Mark, 2001; Goldberger et al., 2000), also from PhysioNet. The database contains 25 records of approximately 10 hours each, sampled at 250 Hz, with rhythm annotations indicating segments of normal sinus rhythm and segments of atrial fibrillation. Of the 25 records, 23 were processed successfully and 2 produced sampling errors during loading. The strict rhythm-labeling policy used here requires that a window fall fully within a single rhythm segment, yielding 11,988 evaluable 60-second windows.

CHB-MIT Scalp EEG Database. The neural tests use the CHB-MIT Scalp EEG Database (Shoeb, 2009; Goldberger et al., 2000) from PhysioNet. The database contains continuous scalp EEG recordings from pediatric epilepsy patients, sampled at 256 Hz across 23 EEG channels. Seizure onsets and ends are provided in the dataset annotations. The preictal window for each seizure is defined here as the period from 30 minutes before seizure onset to 5 minutes before seizure onset, following common practice in seizure prediction literature (Shoeb, 2009). The interictal period excludes the postictal 30 minutes following each seizure. The patients evaluated in this paper are chb01, chb02, chb03, and chb04. The single-patient K-of-N event-detection result is reported on chb01 (42 EDF files, 7 annotated seizures).

3.8 Audit Protocol

Each test writes structured output artifacts and audit information sufficient to inspect the run configuration, the computed window values, and the summary metrics. Where available, run manifests record the dataset path, the record list, the window length, the minimum RR interval count, the coupling constant α, the entropy binning rule, the threshold scope, the target false-positive rate, software version identifiers, package versions, the operating system, and the timestamp of execution. The execution stage runs the analysis on real signal data streamed from PhysioNet (cardiac) or read from local EDF files (neural) without synthetic data substitution, and writes per-window output to CSV files including the timestamp, the computed I, ρ, S, and Φ values, the per-record threshold, the binary instability flag, and the ground-truth label.

The audit stage applies a set of sanity checks across the validation suite, including, where applicable: a shuffle test in which labels are randomly permuted and AUC is recomputed (expected value approximately 0.50); a global threshold comparison in which a single threshold is applied across all records, used to detect whether binary operating metrics are being inflated by per-record calibration; an alpha sensitivity check in which α is varied and AUC is recomputed (expected to move smoothly under perturbation rather than collapse or jump discontinuously); and a manual spot check on a single record's per-window output. For event-level neural tests, two additional audit tests are run: the 3-consecutive-hit test verifies that event-level performance does not depend on the loose single-hit criterion, and the K-of-N test verifies that the K-of-N rule produces event predictions consistent with the per-window data. The audit reports for each test are stored in a parallel directory structure to the result outputs.

3.9 Reference Implementation

The reference implementation of the metric and the test scripts is in the public repository at `https://github.com/Wise314/phi-bio-stability`. The cardiac test script is `scripts/test_phi_ecg_mitdb.py`. The atrial fibrillation test script is `scripts/test_phi_ecg_afdb_rhythm.py`. The neural test script is `scripts/test_phi_eeg_chbmit.py`. The HRV comparison script is `scripts/test_phi_vs_hrv_mitdb.py`. The alpha optimization scripts are `scripts/optimize_alpha_cardiac.py` and `scripts/optimize_alpha_auc.py`. Implementation dependencies include WFDB 4.3.0, antropy 0.1.9, NumPy, SciPy, pandas, and pyEDFlib.

3.10 Scope and Boundary Conditions

The protocol evaluates Φ under per-record (cardiac) and per-patient (neural) baseline threshold calibration on publicly available cardiac and neural physiological datasets. Each test fixes the formula structure, the observable mappings, and the coupling constant before execution for that test, and reports the resulting Φ values, AUC values, and event-level metrics. This paper evaluates retrospective analysis on annotated PhysioNet datasets rather than prospective deployment monitoring. The protocol does not produce a clinical pass-or-fail classification; window-level Φ and event-level detection are reported as continuous and binary measurements respectively, and clinical interpretation is left to subsequent work.

Results reported in this paper are bounded to the specific datasets, observable mappings, configurations, and audit protocols used. Generalization beyond these specific choices, including to other cardiac patient populations, to other physiological modalities, to multi-patient validated seizure prediction, or to deployment on actual wearable hardware, is not established.

4 Baseline Validation

Before reporting the core findings, the protocol's behavior under a null condition was checked to confirm that the per-record threshold calibration and the audit pipeline do not produce spurious AUC inflation. The shuffle audit on the primary cardiac test (Test 1) randomly permuted the NORMAL-compatible and ARRHYTHMIA labels across the same 1,022 evaluable 60-second windows from 34 records and recomputed AUC using the same Φ-derived scores and evaluation code. The shuffle audit produced AUC 0.5003, within 0.0003 of the chance value of 0.5000. The same shuffle audit was applied to the AFib test (Test 2) and produced AUC 0.4998, within 0.0002 of chance. The same shuffle audit was applied to the resolution scaling test at 30-second windows (Test 6) and produced AUC 0.5022, within 0.0022 of chance. These shuffle-audit values support the interpretation that the reported AUC values are not explained by label leakage or a trivial evaluation-pipeline artifact.

5 Core Findings

5.1 Cardiac Arrhythmia Detection on MIT-BIH

The primary cardiac test (Test 1) evaluates whether Φ discriminates ARRHYTHMIA from NORMAL-compatible windows on the MIT-BIH Arrhythmia Database under per-record threshold calibration. The test ran on 34 of 48 records that satisfied the inclusion criterion of at least one NORMAL-compatible window for per-record calibration, producing 1,022 evaluable 60-second windows.

The measured values are shown in Table 1. Φ produced AUC 0.9148 on the binary discrimination task. At the per-record calibrated threshold (target false-positive rate 0.01 per record), the binary classification produced precision 91.71%, recall 72.02%, specificity 94.65%, F1 score 0.8068, and Matthews correlation coefficient 0.6936. The confusion matrix counts were TP 332, FP 30, TN 531, FN 129. The shuffle audit produced AUC 0.5003, confirming the result was not an artifact of label leakage or evaluation-pipeline structure.

Quantity	Value
AUC	0.9148
Precision	91.71%
Recall	72.02%
Specificity	94.65%
F1 score	0.8068
MCC	0.6936
TP / FP / TN / FN	332 / 30 / 531 / 129
n (windows)	1,022
Records included	34 of 48
Shuffle AUC	0.5003

Table 1: Test 1 measured values. Cardiac arrhythmia detection on MIT-BIH Arrhythmia Database, 60-second windows, per-record threshold calibration at target false-positive rate 0.01.

The 14 excluded records had no NORMAL-compatible windows under the strict labeling policy and therefore could not be calibrated under the per-record approach used here. This exclusion is a deployment-bound condition rather than a method failure: per-record calibration requires identifiable baseline periods, and patients with continuous arrhythmia at the time of recording do not provide them in this dataset.

5.2 Atrial Fibrillation as a Stable-State Boundary

The atrial fibrillation test (Test 2) evaluates whether Φ classifies AFib against normal sinus rhythm windows under the same formula, the same coupling constant $\alpha = 0.1$, and the same per-record threshold calibration as Test 1. The test ran on 23 of 25 records on the MIT-BIH Atrial Fibrillation Database, producing 11,988 evaluable 60-second windows under the strict policy that each window fall fully within a single rhythm segment.

The measured values are shown in Table 2. Φ produced AUC 0.5556, near the chance value of 0.5000. At the per-record calibrated threshold, the binary classification produced precision 66.02%, recall 4.20%, specificity 98.89%, F1 score 0.0790, and MCC 0.1006. The confusion matrix counts were TP 171, FP 88, TN 7,829, FN 3,900. The shuffle audit produced AUC 0.4998, supporting the interpretation that the near-chance AFib result was not caused by label leakage or an implementation artifact. An alpha-sensitivity check at $\alpha = 0.05$ produced AUC 0.5559 and at $\alpha = 0.20$ produced AUC 0.5545, confirming that the chance-level result was not driven by a poor choice of coupling constant.

Quantity	Value
AUC	0.5556
Precision	66.02%
Recall	4.20%
Specificity	98.89%
F1 score	0.0790
MCC	0.1006
TP / FP / TN / FN	171 / 88 / 7,829 / 3,900
n (windows)	11,988
Records included	23 of 25
Shuffle AUC	0.4998
AUC at $\alpha = 0.05$	0.5559
AUC at $\alpha = 0.20$	0.5545

Table 2: Test 2 measured values. Atrial fibrillation classification on MIT-BIH Atrial Fibrillation Database, 60-second windows, per-record threshold calibration at target false-positive rate 0.01.

The chance-level AUC on AFib classification is reported as a structural boundary on the metric's scope rather than as a method failure. The Φ formulation evaluates instability against a baseline-referenced quality measure and a temporal coherence measure; both quantities may become less informative once a sustained AFib segment is treated as a stable rhythm class rather than as an onset or transition event. The same metric that discriminates arrhythmia-associated windows from NORMAL windows at AUC 0.9148 does not discriminate sustained AFib from sustained normal sinus rhythm in this configuration.

5.3 Resolution Scaling Across Window Lengths

The resolution scaling test (Test 6) evaluates whether Φ degrades gracefully as the analysis window shortens from clinical-style 60-second windows toward shorter low-latency windows. The test ran on the same MIT-BIH Arrhythmia data and the same per-record threshold-calibration procedure as Test 1, varying the window length and associated minimum-RR requirement. The window lengths evaluated were 60, 30, 10, and 5 seconds.

The measured values are shown in Table 3. AUC degrades smoothly from 0.9148 at 60 seconds to 0.6376 at 5 seconds. Precision remains at or above 61.9% across all four window lengths. Recall degrades faster than precision, falling from 72.0% at 60 seconds to 4.7% at 5 seconds. The shuffle audit at 30-second windows produced AUC 0.5022, confirming that the sub-clinical-resolution AUC was not a calibration artifact.

Window length	AUC	Precision	Recall	n (windows)
60 s	0.9148	91.7%	72.0%	1,022
30 s	0.8775	93.4%	53.8%	2,285
10 s	0.7642	92.2%	32.1%	7,604
5 s	0.6376	61.9%	4.7%	15,975

Table 3: Test 6 measured values. Resolution scaling on MIT-BIH Arrhythmia data across four window lengths, same per-record threshold-calibration procedure as Test 1.

The smooth degradation profile is consistent with a sample-size limit on the metric's three components. At longer window lengths, each window contains enough RR intervals to provide sample support for the standard-deviation-based identity component, the lag-1 autocorrelation, and the entropy histogram. At 5 seconds, each window contains only approximately 3 to 6 heartbeats, which limits the statistical reliability of all three components simultaneously. Precision remains above 90% at 10- to 30-second windows but falls to 61.9% at 5 seconds. Recall drops sharply across resolutions because fewer arrhythmic windows produce Φ values below the per-resolution calibrated threshold under the smaller sample.

5.4 Comparison Against Heart-Rate-Variability Metrics

The HRV comparison test (Test 5) evaluates how Φ performs against 16 standard heart-rate-variability metrics on the same MIT-BIH cardiac arrhythmia detection task under matched per-record calibration conditions. The test used the stricter inclusion rule of at least five NORMAL-compatible windows per record, yielding 811 matched evaluable windows.

The measured values are shown in Table 4. The top three HRV metrics by AUC were RMSSD (0.9706), SDSD (0.9705), and SD1 (0.9705). The identity component I alone produced AUC 0.9377, ranked fourth. SDNN produced AUC 0.9258, ranked fifth. The full Φ metric produced AUC 0.9015, ranked sixth, 0.0691 AUC below RMSSD. Shuffle audits on Φ, RMSSD, and SDNN produced AUC 0.5025, 0.5026, and 0.5009 respectively, all near chance, confirming no label leakage in the comparison pipeline.

Rank	Metric	AUC	Type
1	RMSSD	0.9706	Time-domain HRV
2	SDSD	0.9705	Time-domain HRV
3	SD1	0.9705	Nonlinear HRV (Poincaré)
4	I alone (Φ component)	0.9377	Cross-domain component
5	SDNN	0.9258	Time-domain HRV
6	Φ (full metric)	0.9015	Cross-domain stability metric
7	SD1/SD2	0.8930	Nonlinear HRV (Poincaré)
8	HF Power	0.8537	Frequency-domain HRV
9	SD2	0.8483	Nonlinear HRV (Poincaré)
10	ρ alone (Φ component)	0.8477	Cross-domain component

Table 4: Test 5 top ten metrics by AUC on MIT-BIH cardiac arrhythmia detection. 811 matched evaluable windows under stricter inclusion rule (at least five NORMAL-compatible windows per record). Φ achieves AUC 0.9015, 0.0691 AUC below the best cardiac-specific metric (RMSSD).

The Φ AUC of 0.9015 on Test 5 is slightly below the Φ AUC of 0.9148 from Test 1, reflecting the stricter inclusion rule used in Test 5. The 0.0691 AUC gap between Φ and the best cardiac-specific HRV metric is reported as the empirical cost of cross-domain applicability on this specific cardiac task: Φ trades approximately 0.07 AUC for the property that the same equation structure applies across mechanical, electrical, aerospace, computational, geophysical, quantum, and now biological signals, with $\alpha = 0.1$ retained for the cardiac tests reported here. The result that the identity component I alone outperforms the full Φ metric on this cardiac task (0.9377 vs 0.9015) is reported as an honest internal observation: the entropy term $\alpha \times S$ slightly reduces cardiac-specific accuracy under this configuration, while being retained to preserve the three-component cross-domain formulation evaluated in the broader portfolio.

5.5 Modality- and Patient-Dependent Coupling Constant

The coupling constant grid search compares cardiac and neural alpha-sensitivity profiles. The cardiac alpha grid was computed by the cardiac alpha-optimization script (`scripts/optimize_alpha_cardiac.py`) on the saved Test 1 MIT-BIH window-level outputs, using that script's evaluation pipeline; the resulting AUC values are reported as a sensitivity profile and are not the primary Test 1 summary AUC of 0.9148 reported in Table 1. Under this evaluation pipeline, AUC was nearly flat across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$, with values of 0.9022, 0.9022, 0.9012, and 0.8995 respectively. On CHB-MIT chb01 EEG data with direction pre-specified as "lower Φ = preictal," AUC increased from 0.5409 at $\alpha = 0.10$ to a shallow maximum of 0.5808 at $\alpha = 0.55$, with nearby values 0.5807 at $\alpha = 0.50$ and 0.5806 at $\alpha = 0.60$. A direct comparison on chb04 with all other parameters held constant produced AUC 0.37 at $\alpha = 0.10$ and AUC 0.61 at $\alpha = 0.55$.

The cardiac alpha-sensitivity profile supports the use of $\alpha = 0.1$ as a stable cardiac default, consistent with the most common value used in the author's prior cross-domain evaluations. The EEG profile shows that under the lower-direction interpretation, the AUC improvement from $\alpha = 0.1$ to $\alpha = 0.55$ on chb01 is approximately 0.04 AUC, and the improvement on chb04 is approximately 0.24 AUC. The neural alpha-sensitivity profile does not generalize across all four tested EEG patients: chb02 produces AUC

0.31 at $\alpha = 0.55$ under the chb01-direction convention (opposite to the chb01 and chb04 direction), and chb03 produces AUC 0.51 in either direction. The reported finding is that α may be modality- and patient-dependent rather than fixed across all EEG patients, with cardiac stable at $\alpha = 0.1$ and neural exploratory at $\alpha = 0.2$ (single-patient K-of-N) or $\alpha = 0.55$ (multi-patient optimization on lower-direction patients).

5.6 Self-Falsification of the Original EEG Event Result

The methodological self-falsification (Test 3d) evaluates whether the original Test 3 reported 71% event-level seizure sensitivity on chb01 (5 of 7) was a real signal or a chance accumulation across many independent windows under a permissive single-hit criterion.

The audit ran on the same Φ values and the same per-patient threshold from the original Test 3, changing only the event-detection rule from "at least 1 preictal hit anywhere in the preictal window" to "at least 3 consecutive preictal predictions." Under the 3-consecutive rule, event sensitivity dropped from 71.43% (5 of 7) to 0% (0 of 7). Window-level AUC was unchanged at 0.5654 across both audit framings, confirming the audit was a change in the event-level rule and not a change in the underlying window-level signal.

Two 3-consecutive audit framings are reported in this paper. Test 3d, run at target false-positive rate 0.01 on the original Test 3 configuration, produced 0 of 7 detected seizures under the 3-consecutive rule. A subsequent comparison run, Test 3f, was performed at target false-positive rate 0.05 in the K-of-N evaluation configuration described in Section 5.7 and produced 1 of 7 detected seizures under the same 3-consecutive rule; this later framing is reported as the baseline column in Table 5. Window-level AUC was 0.5654 in both framings. Both numbers are reported because the 3-consecutive rule is run under different per-window false-positive operating points in the two audits.

A chance-rate calculation accounts for the original 71% number under the assumption of independent windows: with target false-positive rate 0.01 per window and approximately 150 preictal windows per seizure, the probability of at least one chance hit per seizure under the original single-hit rule is $1 - (1 - 0.01)^{150} = 1 - 0.99^{150} \approx 0.78$, or approximately 78%. The observed 71% original sensitivity falls within statistical noise of this 78% chance baseline. The original event-level result was therefore reported as falsified by chance accumulation on a low-AUC window-level signal multiplied across many independent windows per seizure.

5.7 K-of-N Event Detection Under Coverage Guard

The K-of-N test (Test 3f) evaluates whether replacing the 3-consecutive rule with a K-of-N rule (3 hits in any 10 consecutive windows, sliding by one window per step) increases event-level sensitivity for scattered threshold-hit patterns without falling back into the chance-accumulation problem of single-hit criteria.

The test ran on chb01 with $\alpha = 0.2$, target false-positive rate 0.05 for the per-patient threshold, refractory period 1,800 seconds, and a coverage guard requiring at least 10 valid preictal windows per seizure. Of seven annotated seizures, six had sufficient preictal coverage and one (chb01_21) had only three preictal windows total and was excluded from the K-of-N evaluation under the coverage guard. The 3-consecutive comparator on the same configuration produced sensitivity 1 of 7 (14%). The K-of-N rule produced sensitivity 4 of 6 (67%) among seizures with sufficient coverage, or 4 of 7 (57%) when the excluded seizure is counted as a miss. Event-level false alarm rate was 0.74 per hour. Window-level AUC was unchanged at 0.5654.

Quantity	3-consecutive rule	K-of-N rule
Sensitivity (all seizures)	1 of 7 (14%)	4 of 7 (57%)
Sensitivity (K-of-N coverage set)	—	4 of 6 (67%)
False alarms per hour	—	0.74
Window-level AUC	0.5654	0.5654
α	0.2	0.2
Target FPR per window	0.05	0.05
Refractory period	1,800 s	1,800 s
Coverage guard (min preictal windows)	—	10

Table 5: Test 3f measured values. K-of-N event detection vs 3-consecutive on chb01 under matched configuration. Single-patient exploratory result.

The K-of-N improvement over the 3-consecutive rule is interpreted as improved sensitivity to scattered threshold-hit patterns rather than clustered runs. Per-seizure breakdown on chb01 shows the mechanism: chb01_15 had 33 preictal hits with a longest consecutive run of 2 (failing the 3-consecutive rule) but multiple qualifying 3-of-10 blocks. An audit on chb04 (Test 3g) provided a check against K-of-N inflation in that patient: 3-consecutive and K-of-N both detected 1 of 3 chb04 seizures, with the detected seizure (chb04_05) having 12 consecutive preictal hits under the tested configuration. The K-of-N result is reported as single-patient exploratory event-level evidence, not as a validated cross-patient seizure prediction claim.

Multi-patient performance under Test 3g with $\alpha = 0.55$ and direction-specific scoring produced patient-dependent results: chb01 sensitivity 67%, chb02 0%, chb03 50%, chb04 33%. The patient-dependent profile is consistent with the patient-specific direction observation reported in the boundary findings section.

5.8 Calibrated Thresholds in the Unnormalized-Entropy Implementation

This section reports the threshold calibration approach used throughout the physiological tests rather than a head-to-head experimental comparison. The per-record (cardiac) and per-patient (neural) threshold calibration used throughout the physiological tests differs from the fixed $\Phi_c = 0.25$ threshold used in the prior portfolio's normalized-entropy implementations. In the present implementation, the entropy component S is computed in bits rather than normalized to $[0, 1]$. The cardiac RR-distribution entropy ranges approximately from 5 to 6 bits, well outside the $[0, 1]$ range that the fixed $\Phi_c = 0.25$ threshold was calibrated against. Per-record (cardiac) and per-patient (neural) threshold calibration at a target false-positive rate is used in place of the fixed threshold for the physiological tests in this paper.

The per-record cardiac thresholds vary substantially across records in the displayed examples. Record 100 produced calibrated threshold 0.076. Record 101 produced 0.004. Record 103 produced 0.116. Record 105 produced 0.271. Record 106 produced 0.129. The displayed examples include calibrated thresholds both below and near the fixed-implementation $\Phi_c = 0.25$ value, confirming that calibrated thresholds differ materially from a fixed threshold across the records examined. A direct head-to-head test applying fixed $\Phi_c = 0.25$ to the same MIT-BIH cardiac data was not run in this paper; the calibrated-threshold approach is established here by the implementation's choice to use bits-entropy and by the patent specification that codifies both fixed-threshold and calibrated-threshold approaches as configurable embodiments.

6 Discussion

6.1 Returning to the Four Questions

The paper opened with four questions. The results support the following answers, each bounded to the tested settings.

The first question asked whether Φ discriminates arrhythmia from normal sinus rhythm on a publicly available cardiac arrhythmia dataset under per-record threshold calibration, and whether the result survives a shuffle-label audit. Test 1 produced AUC 0.9148 on 1,022 evaluable 60-second windows from 34 records of the MIT-BIH Arrhythmia Database. The shuffle audit produced AUC 0.5003, within 0.0003 of chance, supporting the interpretation that the result is not an artifact of label leakage or evaluation-pipeline structure. The cardiac result is the primary validated finding of the paper.

The second question asked whether Φ classifies atrial fibrillation against normal sinus rhythm on a publicly available atrial fibrillation dataset, and what the result says about the scope of what Φ measures. Test 2 produced AUC 0.5556 on 11,988 evaluable windows from 23 records of the MIT-BIH Atrial Fibrillation Database. The shuffle audit produced AUC 0.4998, and alpha-sensitivity checks at $\alpha = 0.05$ and $\alpha = 0.20$ produced AUC values within 0.001 of the primary result. The chance-level AUC is reported as a structural boundary on the metric's scope: the same formula that discriminates arrhythmia-associated windows from NORMAL windows at AUC 0.9148 does not discriminate sustained AFib from sustained normal sinus rhythm in this configuration.

The third question asked whether Φ on cardiac data degrades gracefully across window lengths, and how Φ compares against standard heart-rate-variability metrics on the same task. The resolution scaling test (Test 6) produced AUC 0.9148, 0.8775, 0.7642, and 0.6376 at window lengths of 60, 30, 10, and 5 seconds respectively, with no abrupt collapse and with precision remaining above 90% at 10- and 30-second windows. The HRV comparison test (Test 5) ranked Φ sixth among 16 standard HRV metrics on the same MIT-BIH cardiac arrhythmia detection task, 0.0691 AUC below the best cardiac-specific metric (RMSSD). The HRV gap is reported as the empirical cost of cross-domain applicability on this specific cardiac task.

The fourth question asked whether the same formula structure transfers to neural electroencephalography signals for seizure prediction, and what the boundary conditions are on that extension. The exploratory neural results produced weak window-level AUC values (0.51 to 0.61 across the four tested patients) with patient-specific direction. The original Test 3 reported 71% event-level seizure sensitivity on chb01 was reduced to 0 of 7 under a 3-consecutive-hit audit, with a chance-rate calculation showing approximately 78% probability of at least one chance hit per seizure under the original single-hit criterion. The K-of-N event-detection rule increased single-patient event sensitivity to 67% (4 of 6 seizures with sufficient coverage) at 0.74 false alarms per hour. The K-of-N result is reported as single-patient exploratory event-level evidence, not as a validated cross-patient seizure prediction claim.

6.2 Cardiac as the Primary Validated Layer

The cardiac result is the strongest finding in the paper. The combination of AUC 0.9148, shuffle audit AUC 0.5003, full confusion matrix counts on 1,022 windows, and the structural boundary established by the AFib negative result on 11,988 windows produces a primary validation that does not depend on a single test or a single audit. The resolution scaling result shows that the same formula and the same coupling constant retain signal across a 12× window-length range from 60 to 5 seconds, with the per-record threshold-calibration procedure recalibrating at each resolution. The HRV comparison establishes that Φ does not outperform cardiac-specific HRV metrics on cardiac-specific tasks; the value of the cross-domain formulation is structural rather than performance-based on any single domain.

6.3 Neural as the Exploratory Layer

The neural result is exploratory throughout. Window-level AUC across the four tested patients is in the 0.51 to 0.61 range, modestly above chance for two patients and inverted for one. Patient-specific direction calibration is required because preictal physiology presents differently across patients: chb01 and chb04 show "lower Φ = preictal" while chb02 shows the opposite. An absolute-deviation variant tested as a direction-agnostic alternative did not recover the lost signal across the tested patients. The exploratory single-patient K-of-N event sensitivity of 67% at 0.74 false alarms per hour on chb01 is the strongest event-level result in the paper, but it remains single-patient evidence under per-patient threshold calibration, $\alpha = 0.2$, and a coverage guard that excluded one of seven seizures from K-of-N evaluation. Multi-patient performance under $\alpha = 0.55$ and direction-specific scoring produced the patient-dependent profile of 67%, 0%, 50%, and 33% across chb01, chb02, chb03, and chb04, consistent with the patient-specific direction observation.

The crude observable mapping used in the neural tests, a per-channel z-scored channel-aggregated signal summarized by RMS amplitude, signed autocorrelation, and spectral-entropy deviation, is a deliberate choice to keep the neural mapping structurally simple rather than a competitive seizure-prediction pipeline. The exploratory framing of the neural results in this paper reflects that choice. Future work can test whether richer neural observable mappings, including spectral band power per channel, phase synchrony, or graph-theoretic measures across the full 23-channel recording, improve neural results without changing the formula structure.

6.4 The Methodological Self-Falsification

The reduction of the original 71% event-level seizure sensitivity result on chb01 to 0 of 7 under a 3-consecutive-hit audit, with window-level AUC unchanged at 0.5654, is one of the more useful methodological observations in the paper. The chance-rate calculation accounts for the original number under the assumption of independent windows: with target false-positive rate 0.01 per window and approximately 150 preictal windows per seizure, the probability of at least one chance hit per seizure under a single-hit rule is approximately 0.78, within statistical noise of the observed 0.71. The audit replaced a misleadingly strong-looking event-level number with the underlying near-chance window-level signal it was hiding. The K-of-N rule introduced in response to this audit is the corrected event-level method evaluated here, with the coverage guard and refractory period documented as configurable embodiments in the patent specification corresponding to this work.

The audit pattern is general beyond seizure prediction. Monitoring applications that aggregate many weak window-level predictions into event-level predictions through a single-hit criterion can be vulnerable to chance accumulation when the underlying window-level signal has AUC near 0.50 and many independent windows occur per event. The lesson reported here applies to physiological monitoring, equipment monitoring, and any domain in which weak per-window signals are aggregated into event-level decisions.

6.5 The Trade-Off Quantified

The HRV comparison test quantifies the cost of cross-domain applicability on cardiac data: Φ trails the best cardiac-specific HRV metric by 0.0691 AUC. This is the empirical price paid for the property that the same formula structure applies across mechanical, electrical, aerospace, computational, geophysical, quantum, and biological signals. The trade-off is reported as a real cost rather than as a defense of the cross-domain approach. For a cardiac-only benchmark where maximum AUC on this specific task is the sole objective, the HRV metric would be preferred over Φ based on the present comparison. The cross-domain framework may be most useful in settings where comparable stability monitoring across multiple signal types is desired, or where a baseline stability metric is needed before a mature domain-specific feature pipeline exists.

6.6 Boundary and Negative Findings

The paper's negative results are part of the scientific record and not buried in supplementary material. The AFib boundary at AUC 0.5556 defines the metric's scope as transition detection rather than stable-state classification. The patient-specific direction observation on EEG defines a per-patient calibration requirement rather than a single direction. The absolute-deviation test on EEG changed performance from 0.57 to 0.50 on chb01, from 0.70 to 0.72 on chb02, and from 0.54 to 0.51 on chb03, showing that direction-free scoring did not consistently recover signal across the tested patients. The crude observable mapping limitation on EEG places the neural results in exploratory territory and motivates richer neural observable mappings as future work. These boundary findings strengthen rather than weaken the cardiac result by establishing the conditions under which the metric does and does not produce useful signal.

7 Limitations

The evidence reported in this paper has several limitations that bound the interpretation of the results.

The cardiac validation uses two historical PhysioNet ECG datasets: the MIT-BIH Arrhythmia Database and the MIT-BIH Atrial Fibrillation Database. Whether the cardiac result generalizes to other patient populations, to in-hospital telemetry recordings, to wearable consumer devices, or to non-ambulatory clinical settings is not established by this evidence.

The neural validation uses four patients from the CHB-MIT Scalp EEG Database (chb01, chb02, chb03, chb04). Whether the exploratory neural results generalize to the full CHB-MIT cohort, to adult epilepsy patients, to non-pediatric seizure types, or to other scalp EEG datasets is not established. The single-patient K-of-N event detection result on chb01 is reported as single-patient exploratory evidence, not as a validated cross-patient seizure prediction claim.

The observable mappings used in this paper are simple by design. The cardiac mapping uses RR-interval standard deviation, lag-1 RR autocorrelation, and Shannon entropy of the RR distribution in bits. The neural mapping uses RMS amplitude of a per-channel-normalized, channel-aggregated EEG signal, signed 1.0-second-lag autocorrelation of that aggregated signal, and absolute deviation of spectral entropy from a per-patient baseline. Whether richer observable mappings (spectral band power, phase synchrony, graph-theoretic measures, multi-channel decompositions) would improve the cardiac or neural results without changing the formula structure is not established.

The threshold calibration in this paper is per-record (cardiac) or per-patient (neural) at a target false-positive rate. A direct head-to-head test applying fixed $\Phi_c = 0.25$ to the same cardiac data was not run. The calibrated-threshold approach used here is established by the implementation's choice to use bits-entropy and by the patent specification that codifies both fixed-threshold and calibrated-threshold approaches as configurable embodiments. Whether fixed-threshold and calibrated-threshold approaches produce comparable AUC values on identical data is not directly established by the evidence reported here.

The coupling constant grid search is bounded to $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ for cardiac and $\alpha \in \{0.10, 0.20, 0.50, 0.55, 0.60\}$ for chb01 EEG, with a direct-comparison check on chb04 at $\alpha = 0.10$ and $\alpha = 0.55$. Whether α values outside these ranges, or per-patient α optimization on EEG patients other than chb01 and chb04, would produce stronger results is not established.

The paper does not establish a clinical pass-or-fail threshold for any of the tested datasets. Φ is reported as a continuous measurement and the binary instability flag is reported only at the calibrated Φ_c for each record or patient. Clinical interpretation of any specific Φ value depends on the deployment context and is left to subsequent work.

The audit pipeline includes shuffle tests, global threshold comparisons, alpha-sensitivity checks, and manual spot checks where applicable. Audit coverage varies by test. The four cardiac tests and the EEG self-falsification test have the strongest audit coverage. The exploratory EEG K-of-N tests have audit coverage limited to the 3-consecutive-hit comparison and the chb04 inflation check.

Wall-clock per-window computation cost is not characterized in this paper; deployment on resource-constrained hardware would require profiling that is not performed here.

The EEG preictal horizon was fixed at 30 to 5 minutes before seizure onset, following common practice in the seizure prediction literature. Whether Φ behaves differently under longer or shorter prediction horizons is not established by the evidence reported here.

8 Conclusion

This paper evaluated whether a stability metric requiring no supervised model training, defined as $\Phi = I \times \rho - \alpha \times S$, detects rhythm instability across cardiac and neural physiological signals using domain-appropriate observable mappings. The reported findings are as follows.

First, Φ discriminated arrhythmia from normal sinus rhythm on the MIT-BIH Arrhythmia Database at AUC 0.9148 across 1,022 evaluable 60-second windows from 34 records, with a shuffle audit AUC of 0.5003.

Second, Φ produced AUC 0.5556 on the MIT-BIH Atrial Fibrillation Database across 11,988 windows, defining a structural boundary on the metric's scope: the same formula that detects arrhythmia transitions does not classify between sustained stable rhythm modes.

Third, the same formula and the same coupling constant produced AUC values of 0.9148, 0.8775, 0.7642, and 0.6376 at window lengths of 60, 30, 10, and 5 seconds respectively, with no abrupt collapse and with the per-record threshold-calibration procedure recalibrating at each resolution.

Fourth, in a head-to-head comparison against 16 standard heart-rate-variability metrics on the same MIT-BIH cardiac arrhythmia detection task, Φ ranked sixth among 16 metrics with AUC 0.9015, 0.0691 AUC below the best cardiac-specific HRV metric. The trade-off is reported as the empirical cost of cross-domain applicability on this specific cardiac task.

Fifth, exploratory neural validation on the CHB-MIT Scalp EEG Database produced weak window-level AUC values (0.51 to 0.61 across four tested patients) with patient-specific direction. A 3-consecutive-hit audit on chb01 falsified an earlier reported 71% event-level seizure detection result by reducing detection to 0 of 7 while leaving window-level AUC unchanged at 0.5654, consistent with a chance-rate calculation showing approximately 78% chance accumulation per seizure under the original single-hit criterion. A K-of-N event-detection rule (3 hits in any 10 windows, sliding by one window per step) increased single-patient event sensitivity to 67% (4 of 6 seizures with sufficient coverage) at 0.74 false alarms per hour.

Sixth, per-record (cardiac) and per-patient (neural) threshold calibration at a target false-positive rate was used throughout the physiological tests in place of a fixed $\Phi_c = 0.25$ threshold, because the entropy component is computed in bits rather than normalized to $[0, 1]$ in the present implementation. The calibrated-threshold approach is established here by the implementation's choice to use bits-entropy and by the patent specification that codifies both fixed-threshold and calibrated-threshold approaches as configurable embodiments.

The results are bounded to the tested datasets, observable mappings, configurations, and audit protocols. Generalization beyond these specific choices, including to other cardiac patient populations, to other physiological modalities, to multi-patient validated seizure prediction, or to deployment on actual wearable hardware, is not established by the evidence reported here.

Code Availability

The reference implementation of the stability metric, the validation test scripts, the dataset processing scripts, and repository documentation describing the audit structure are available in the public repository at `https://github.com/Wise314/phi-bio-stability`.

The cardiac test script is `scripts/test_phi_ecg_mitdb.py`. The atrial fibrillation test script is `scripts/test_phi_ecg_afdb_rhythm.py`. The neural test script is `scripts/test_phi_eeg_chbmit.py`. The HRV comparison script is `scripts/test_phi_vs_hrv_mitdb.py`. The alpha optimization scripts are `scripts/optimize_alpha_cardiac.py` and `scripts/optimize_alpha_auc.py`. Implementation dependencies include WFDB 4.3.0, antropy 0.1.9, NumPy, SciPy, pandas, and pyEDFlib.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/978,132 (filed February 9, 2026). No license to implement or commercialize the described methods is granted by this publication.

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